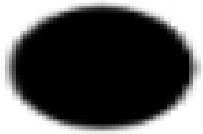
Activity Diagram

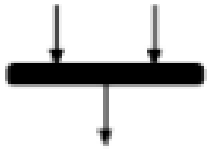
Activity Diagram

An activity diagram is essentially a flowchart that shows activities performed by a system. If you're new to UML diagramming software, this guide will help you get started.

Activity Diagram

Symbol	Name	Description
	Start symbol	Represents the beginning of a process or workflow in an activity diagram. It can be used by itself or with a note symbol that explains the starting point.
	Activity symbol	Indicates the activities that make up a modeled process. These symbols, which include short descriptions within the shape, are the main building blocks of an activity diagram.

Activity Diagram

	Connector symbol	Shows the directional flow, or control flow, of the activity. An incoming arrow starts a step of an activity; once the step is completed, the flow continues with the outgoing arrow.
	Joint symbol/ Synchronization bar	Combines two concurrent activities and re-introduces them to a flow where only one activity occurs at a time. Represented with a thick vertical or horizontal line.

Identifying Use Cases



Decision symbol

Represents a decision and always has at least two paths branching out with condition text to allow users to view options. This symbol represents the branching or merging of various flows with the symbol acting as a frame or container.



End symbol

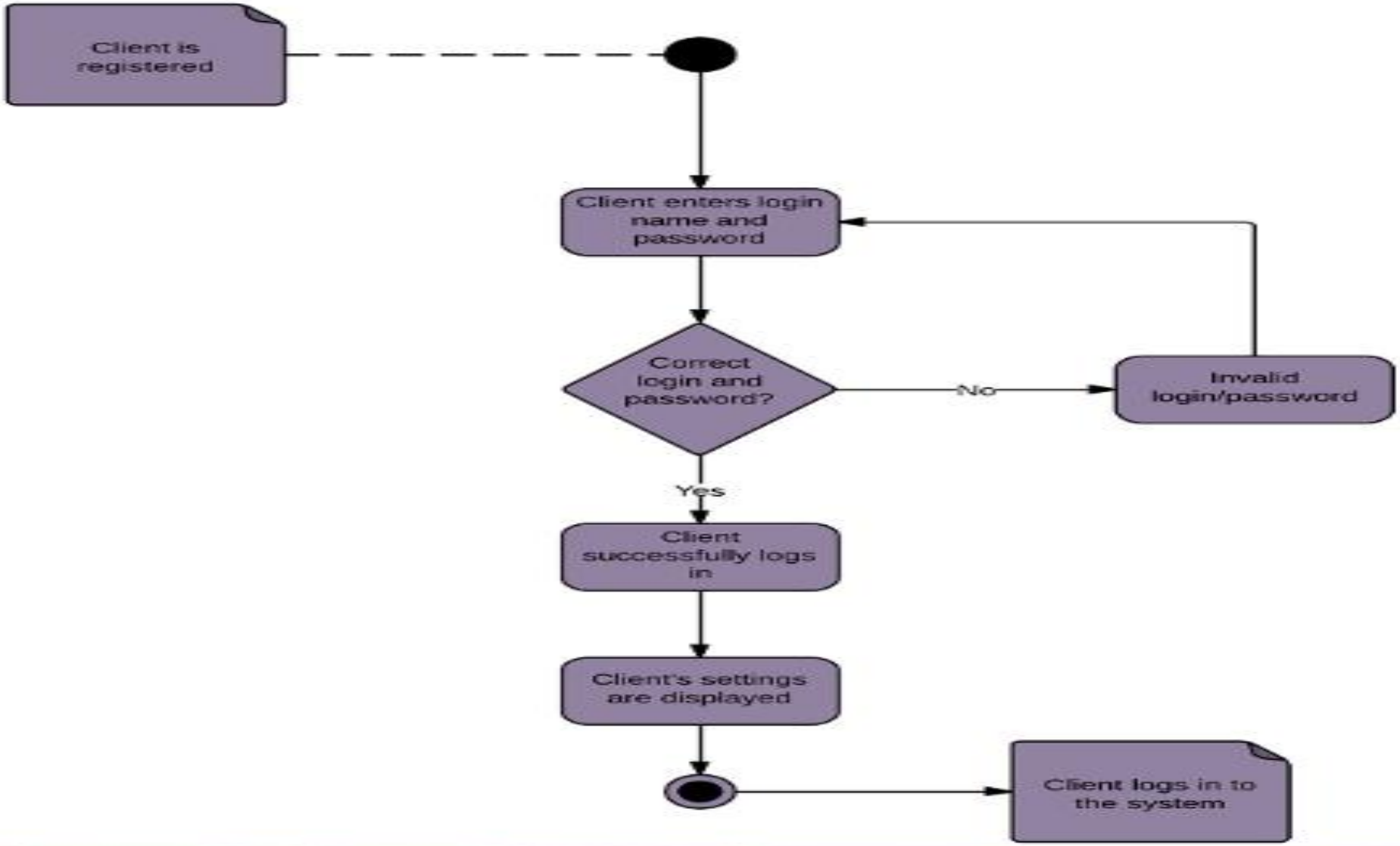
Marks the end state of an activity and represents the completion of all flows of a process.

Activity diagram for a login page

Activity diagram for a login page

Many of the activities people want to accomplish online—checking email, managing finances, ordering clothes, etc.—require them to log into a website. This activity diagram shows the process of logging into a website, from entering a username and password to successfully logging in to the system. It uses different container shapes for activities, decisions, and notes. Lucidchart is the ideal tool for creating any kind of UML flowchart, whether it’s an activity diagram, a use case diagram, or a component diagram. Lucidchart offers in-editor collaboration tools and instant web publishing so you can demonstrate the functionality of your system to others.

Activity diagram for a login page

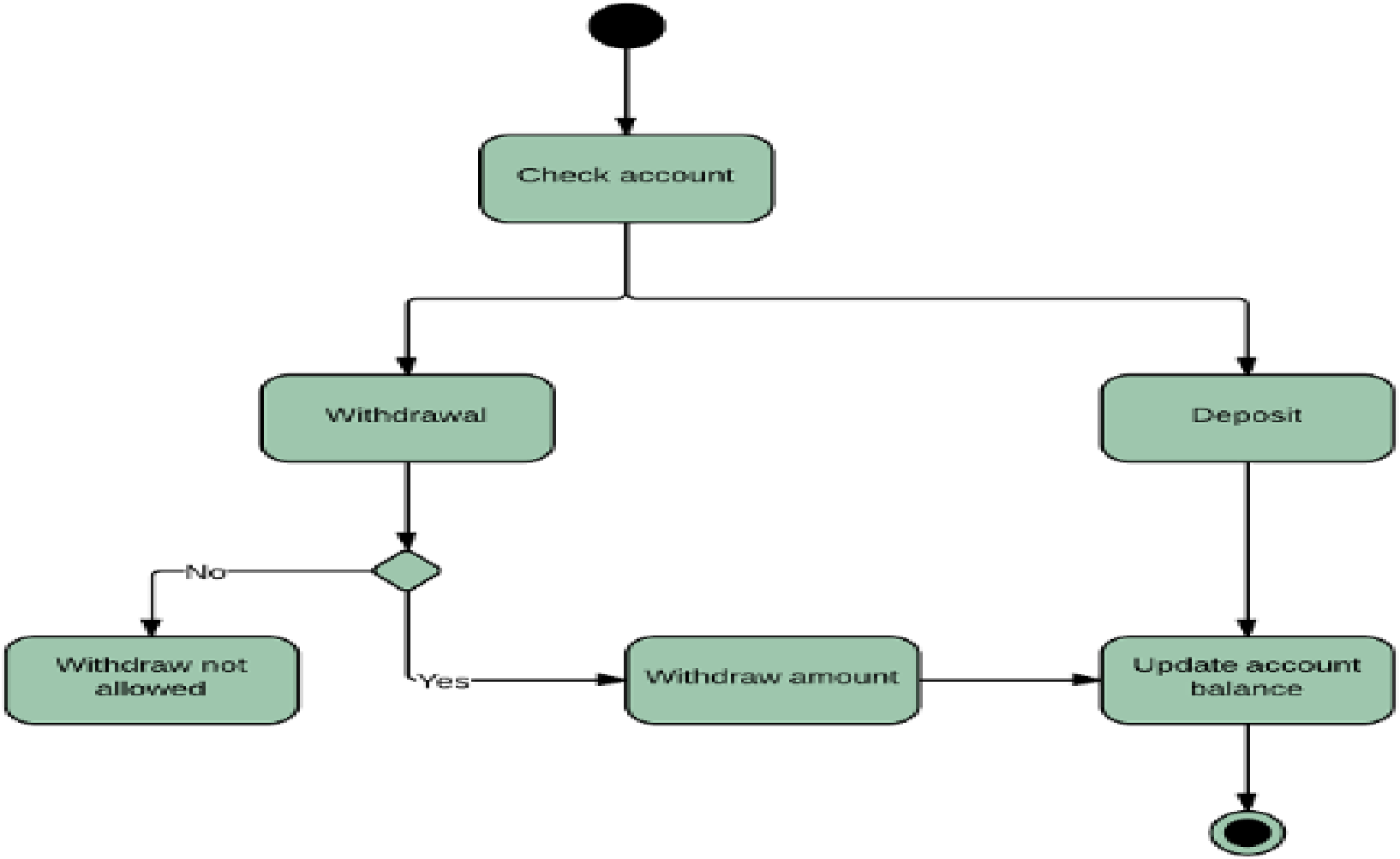


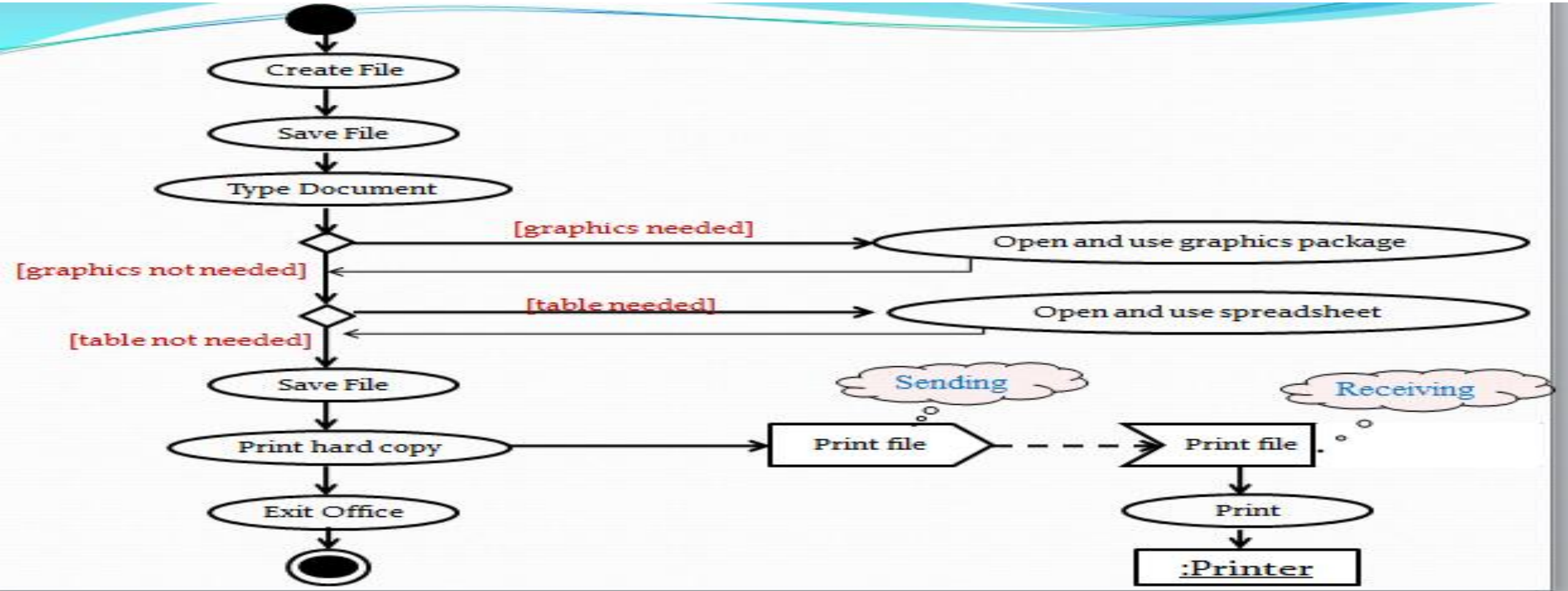
Activity diagram for a banking system

Activity diagram for a banking system

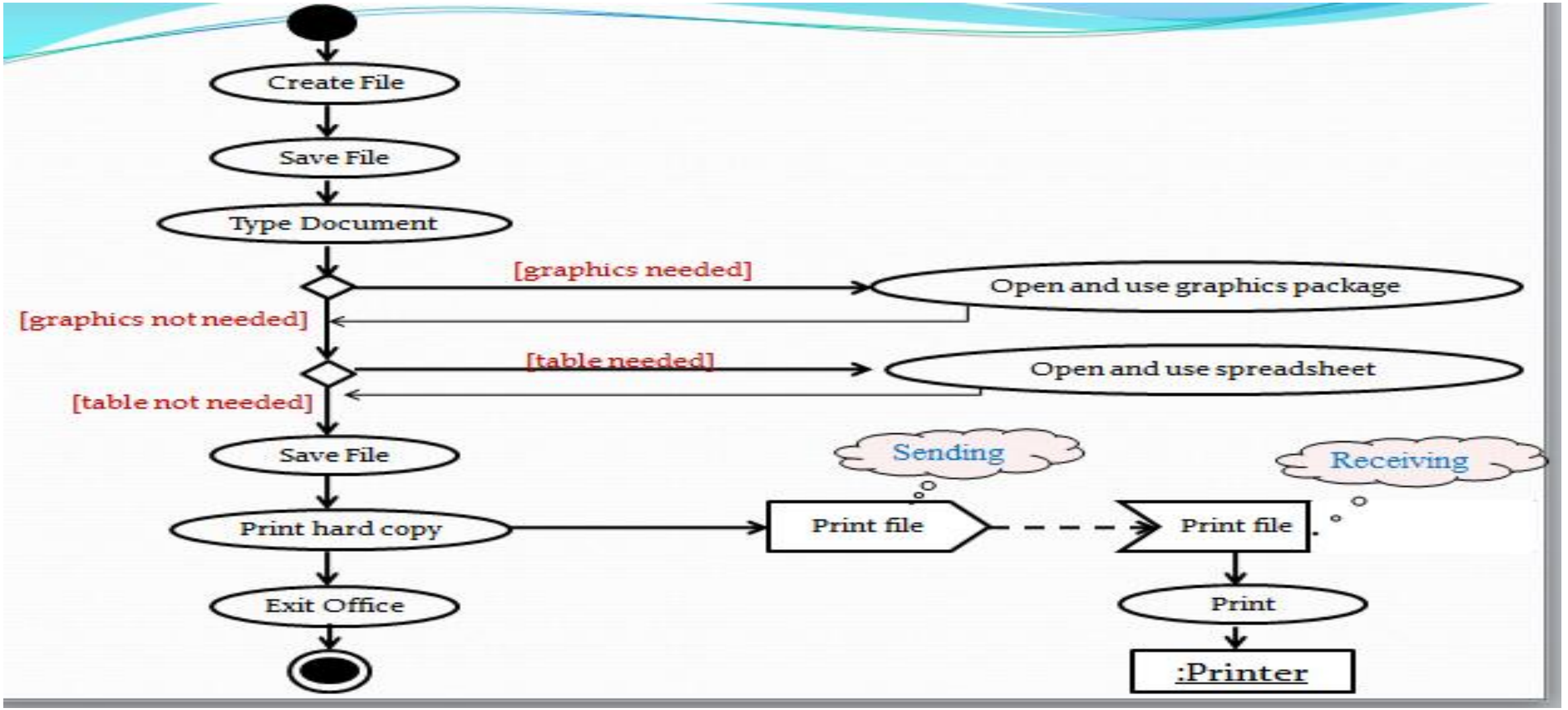
- This diagram shows the process of either withdrawing money from or depositing money into a bank account. An advantage of representing the workflow visually in UML is the ability to show withdrawals and deposits on one chart. When you use Lucidchart to build an activity diagram, you can customize your templates with professional-grade typefaces and colors. Never worry about losing your documents with secure, cloud-based storage.

Activity Diagram for Banking System




Create File
Save File
Type Document
[graphics needed]
Open and use graphics package
[graphics not needed]
[table needed]
Open and use spreadsheet
[table not needed]
Save File
Print hard copy
Exit Office
Sending
Receiving
Print file
Print file
Print
:Printer

Word editing Activity Diagram



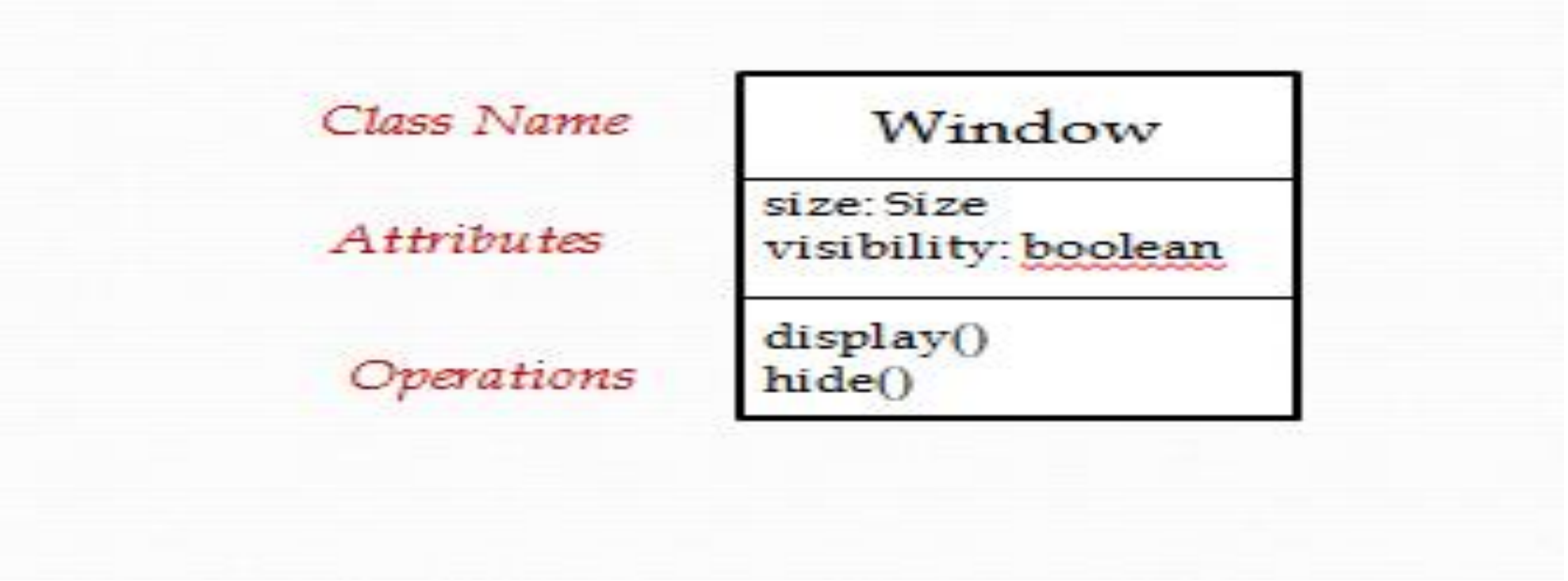
Class Diagram

Elements of a UML Class Diagram

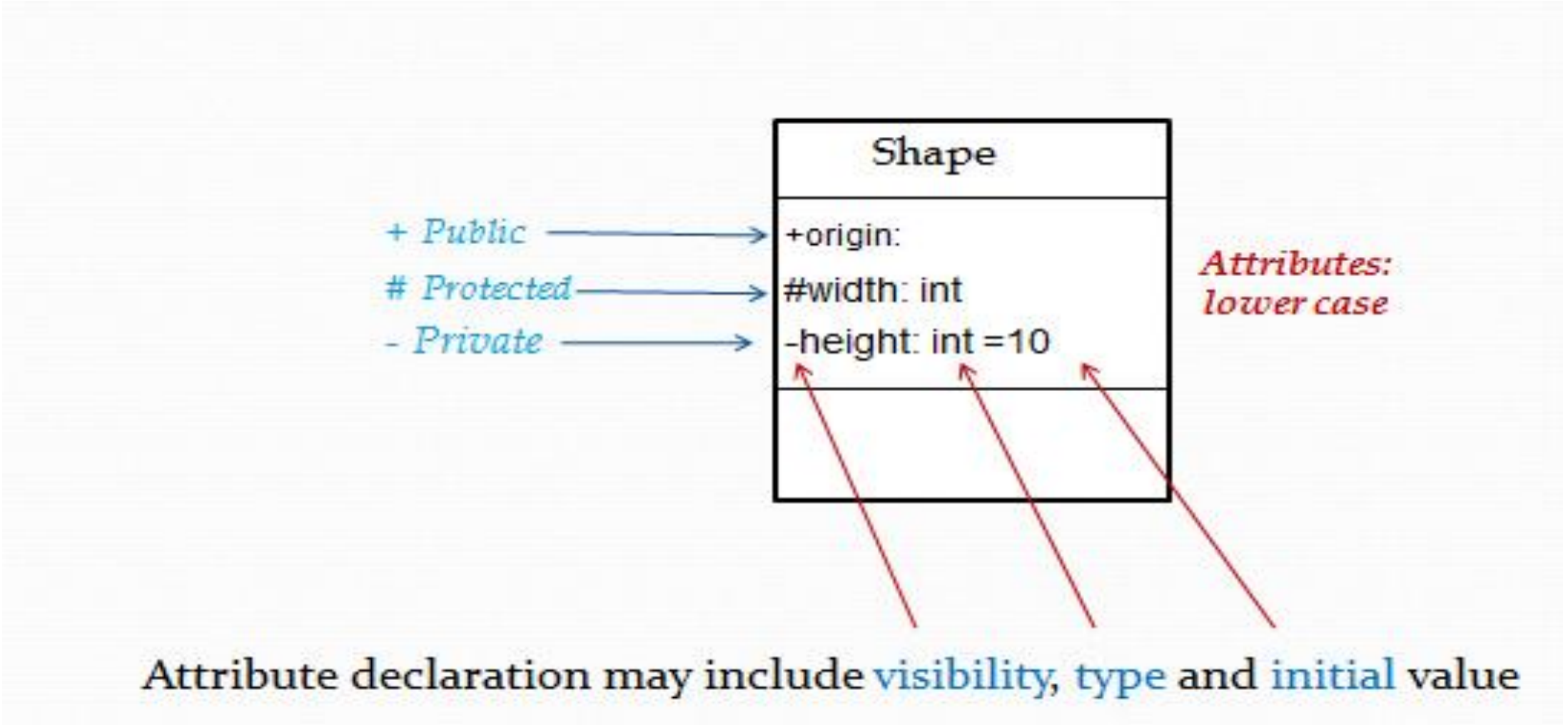
- Class
- Attributes
- Operations
- Relationships

Classes

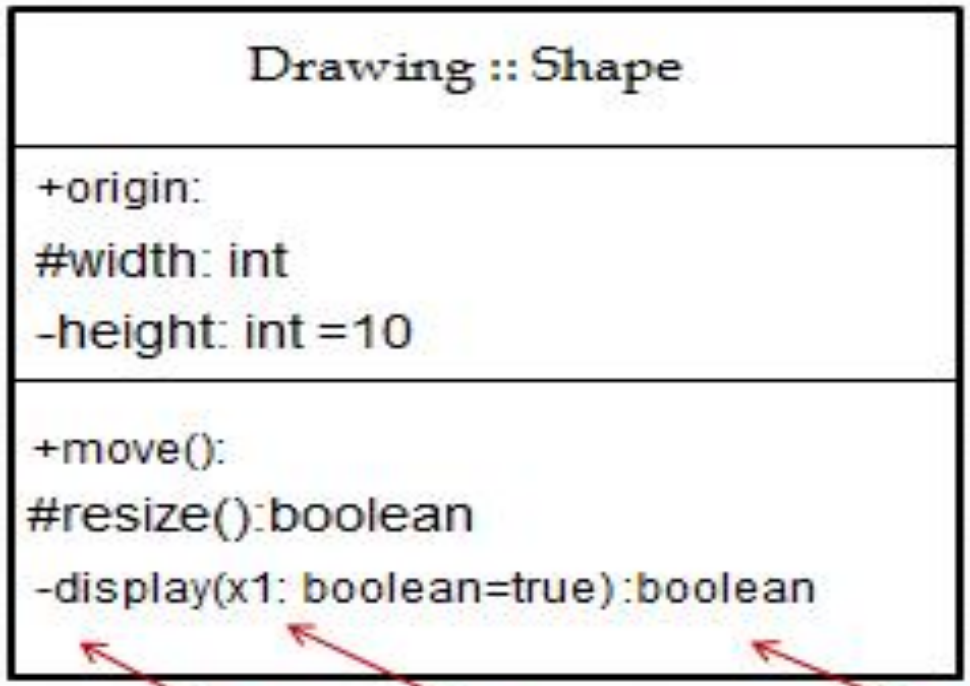
- A class is the description of a set of objects having similar attributes, operations, relationships and behavior.



Class Names And Attributes



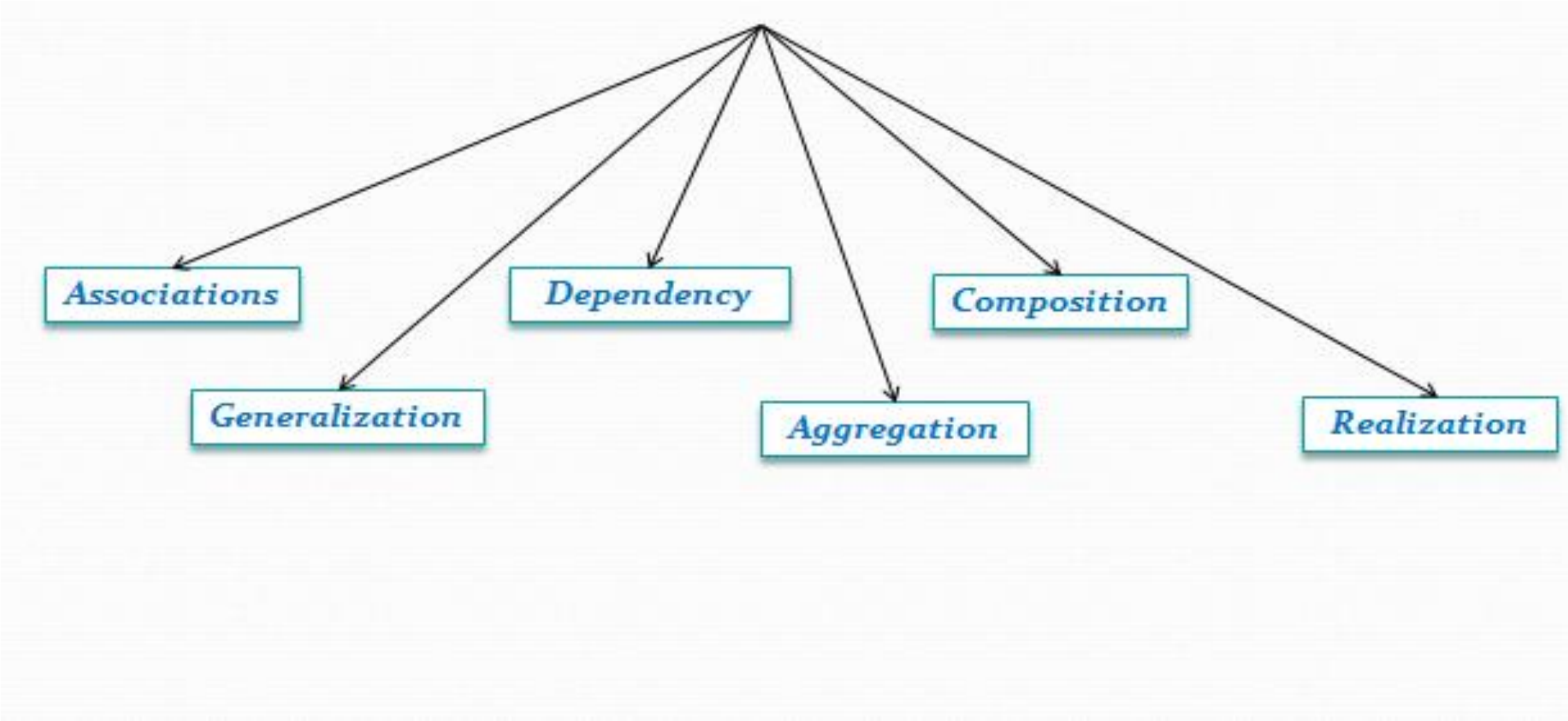
Class Operations



Operations: lower case

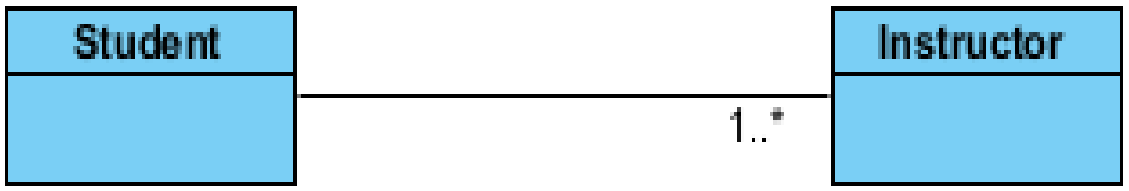
- Operation may include visibility, parameters, and return type:
  +opName(param1 : type = initial_value) : return-type

Relationships



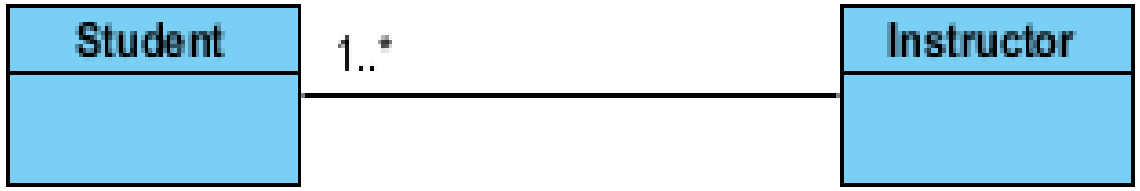
Association and multiplicity

- If two classes in a model need to communicate with each other, there must be link between them, and that can be represented by an association (connector). Association can be represented by a line between these classes with an arrow indicating the navigation direction. In case arrow is on the both sides, association has bidirectional association.
- A single student can associate with multiple teachers:



Association and multiplicity

- The example indicates that every Instructor has one or more Students:



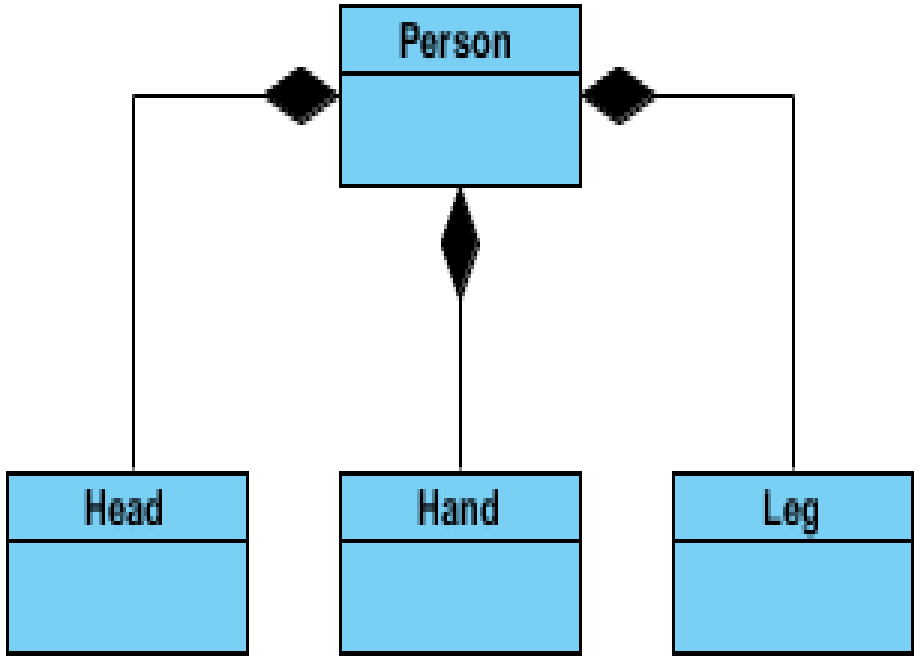
We can also indicate the behavior of an object in an association (i.e., the role of an object) using role names.

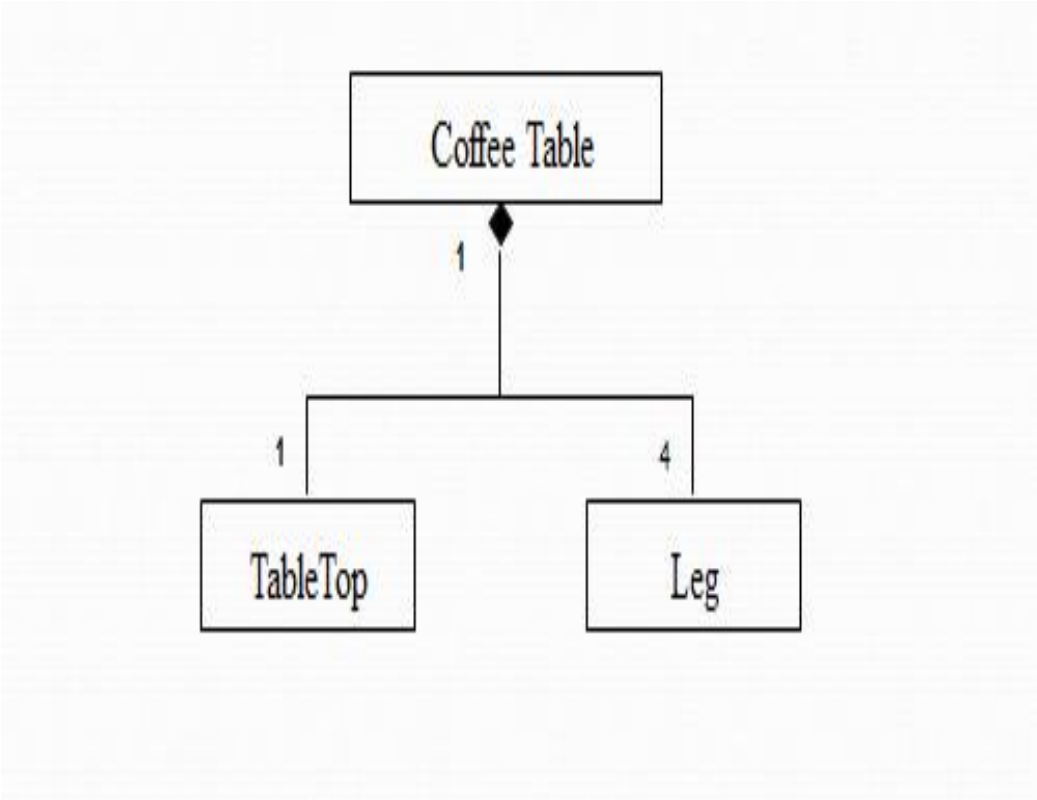


Association vs Aggregation vs Composition

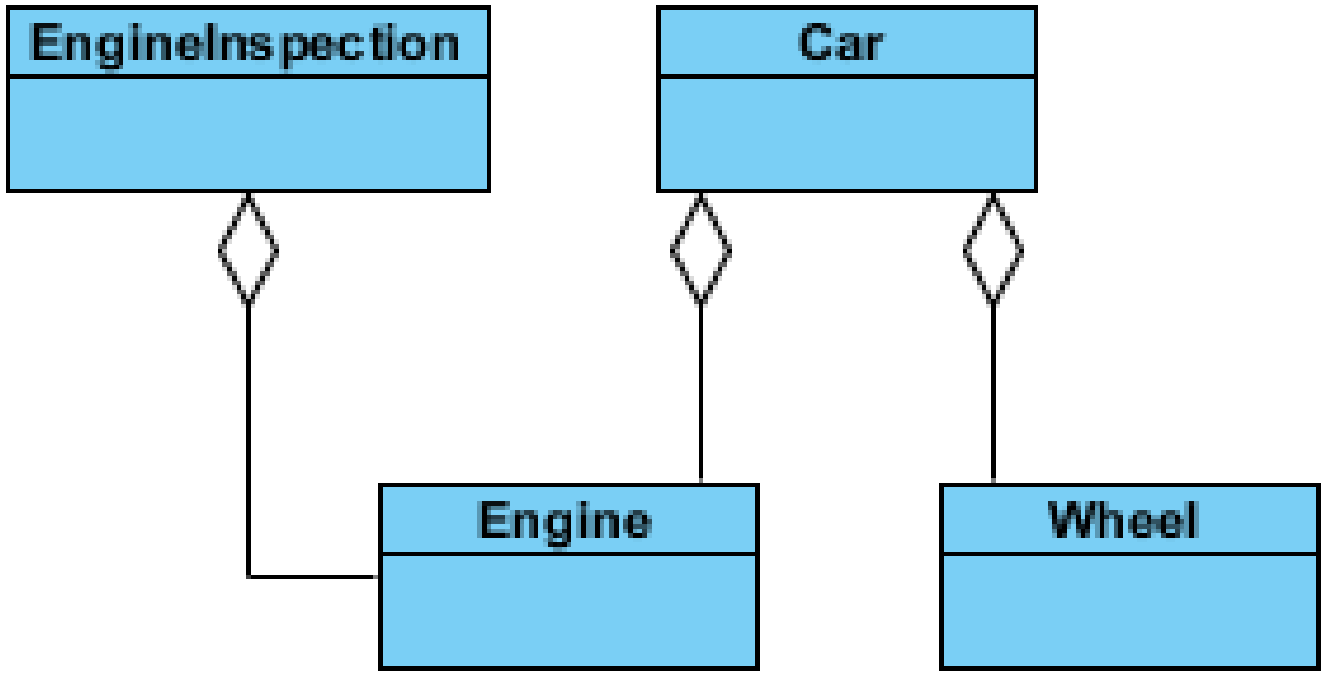
- The question "What is the difference between association, aggregation and composition" has been frequently asked lately.
  Actually, Aggregation and Composition are subsets of association meaning they are specific cases of association. In both aggregation and composition object of one class "owns" object of another class. But there is a subtle difference:
- Aggregation implies a relationship where the child can exist independently of the parent. Example: Class (parent) and Student (child). Delete the Class and the Students still exist.
- Composition implies a relationship where the child cannot exist independent of the parent. Example: House (parent) and Room (child). Rooms don't exist separate to a House.

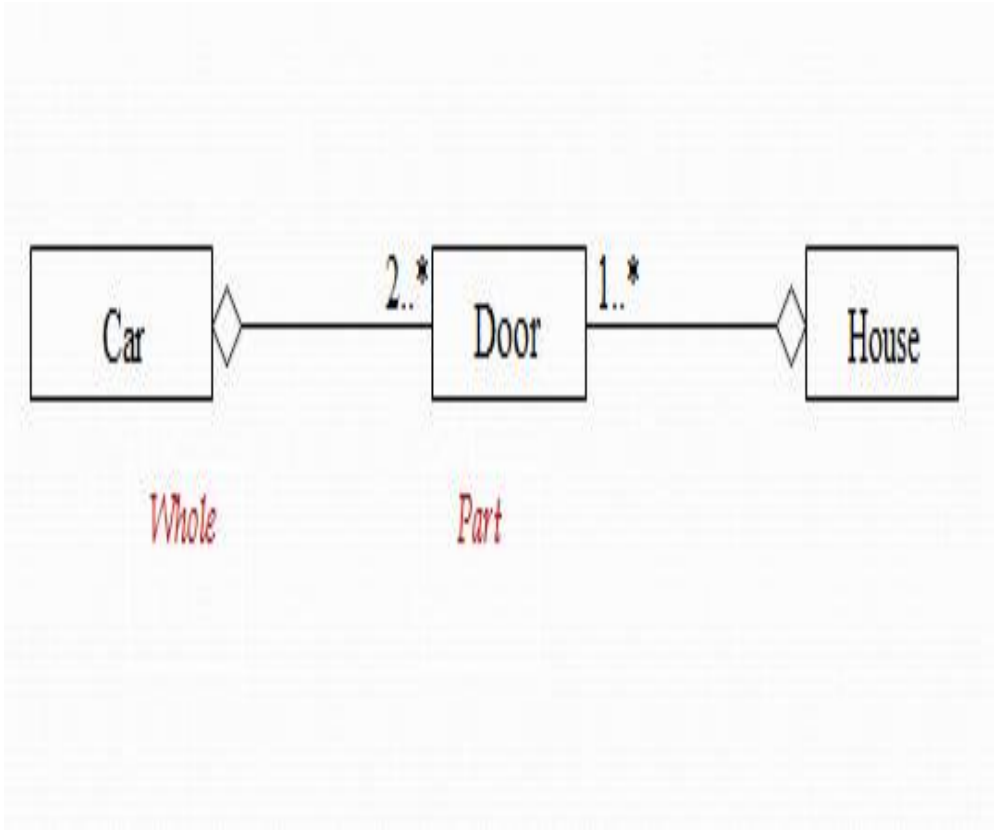
Composition Example:





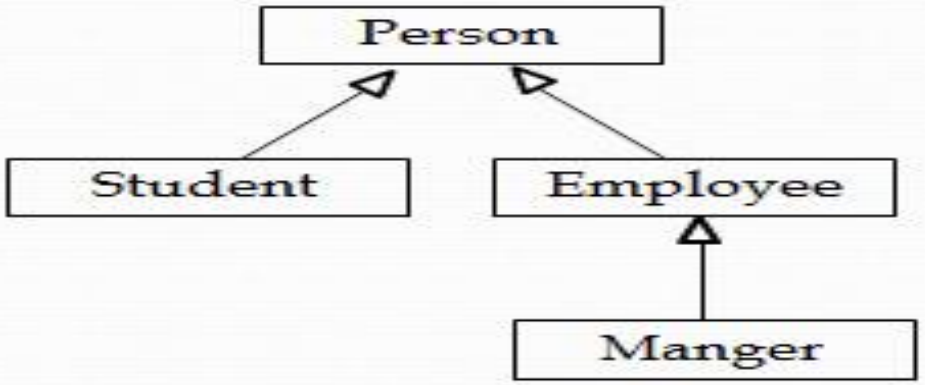
Aggregation Example:





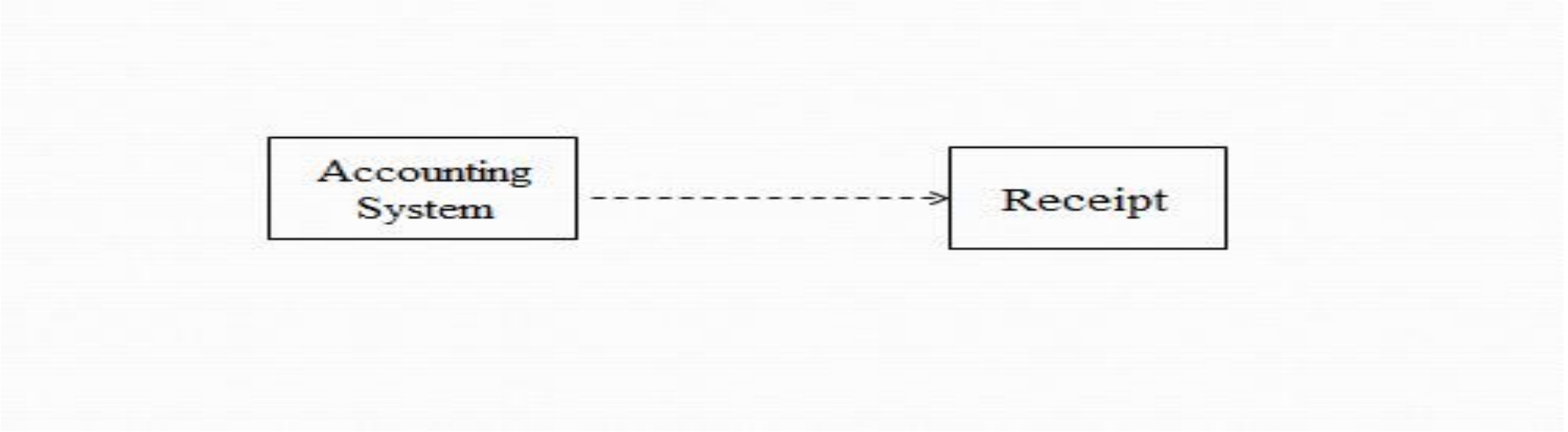
Generalization

- A class might have no parents, in which case it's a base class or root class.
- A class might have no children, in which case it's a leaf class.
- If a class has exactly one parent, it has single inheritance.
- If a class has more than one parent, it has multiple inheritance.



Dependencies

- A kind of relationship, one class uses another. This is called a dependency.



Class Diagram Example

Create class diagrams that describe the classes and relationships depicted in the following scenarios:

- **Researchers are placed into a database that is maintained by the state of Georgia. Information of**

interest includes researcher name, title, position, date began current position, number of years

at current position; university name, location, enrollment; and research interests.

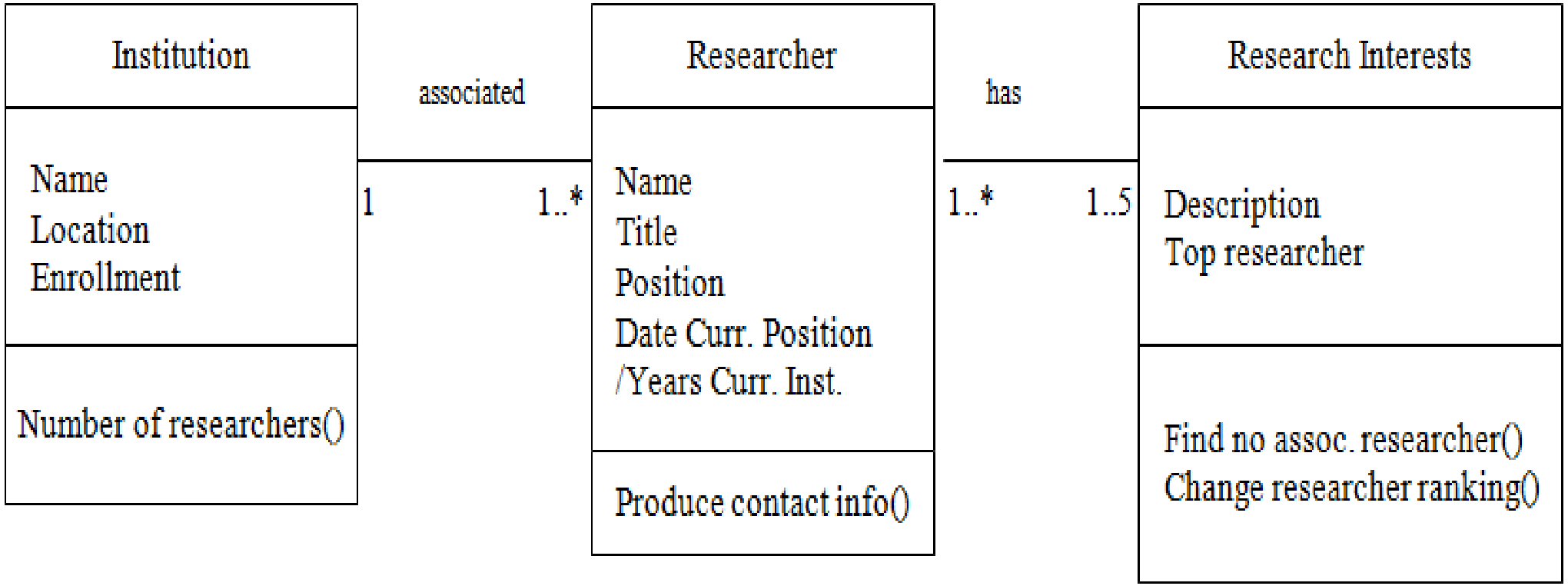
Researchers are associated with one institution, and each researcher can has up to five research interests.

More than one researcher can have the same interest, and the system tracks the ranking of the best researchers for each interest.

The system should be able to insert new researchers, universities, and

research interests; produce information, such as the number of researchers at each university, contact information for the researchers, and research interests that do not have associated researchers; and change researcher rankings

Solution


Institution
Name
Location
Enrollment
Number of researchers()
associated
1
1..*
Researcher
Name
Title
Position
Date Curr. Position
/Years Curr. Inst.
Produce contact info()
has
1..*
1..5
Research Interests
Description
Top researcher
Find no assoc. researcher()
Change researcher ranking()

Thank you